devine-1989-stereotypes-and-prejudice--their-automatic-a

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Paper: devine-1989-stereotypes-and-prejudice--their-automatic-and-controlled-components

Executive Summary

The paper investigates whether high- and low-prejudice individuals differ in their automatic activation of Black stereotypes and in their capacity for controlled inhibition of those stereotypes. It finds that both groups possess equivalent stereotype knowledge and show comparable automatic activation, but only low-prejudice participants demonstrate controlled inhibition, producing more positive, belief-oriented thoughts. The work contributes a dissociation model separating automatic stereotype activation from controlled, nonprejudiced responses.

Reading Recommendation

Read Selectively

The paper offers a theoretically interesting dissociation of automatic and controlled prejudice processes and provides empirical evidence across three experiments, but moderate methodological and reporting weaknesses (misaligned outcome measures in Exp 1, overgeneralized claims about controlled inhibition, incomplete statistical reporting in Exp 2) limit confidence in the robustness of its conclusions. Scholars should focus on the detailed methods and results of Experiments 2 and 3, where the core claims about automatic activation and controlled inhibition are most directly tested.

Background and Motivation

The paper addresses the gap that, despite extensive evidence that cultural stereotypes are automatically activated, it remains unclear how high- and low-prejudice individuals differ--or are similar--in their automatic and controlled responses to target group members, as only a single prior study has directly compared them. This gap is motivated by foundational work on automatic stereotype activation (Dovidio, Evans, & Tyler 1986; Smith & Branscombe 1985) and on the early acquisition and persistence of stereotypes (Allport 1954; Katz 1976).

Theoretical Stance

Overall stance: Consistent Theoretical

The paper introduces the dissociation model in the introduction and repeatedly references it in the discussion, interpreting each experiment's findings as supporting the model's core claims about equal stereotype knowledge, automatic activation, and the need for controlled inhibition. Although the authors occasionally overstate the generality of controlled inhibition, the overall alignment between intro and discussion remains intact.

Theories Discussed

- **Dissociation model of automatic vs. controlled processes in prejudice** *(focal)*: Proposes that stereotype knowledge is equally present in high- and low-prejudice individuals, that stereotypes are automatically activated by target cues, and that nonprejudiced responses require intentional, controlled inhibition of the activated stereotype.
- **Inevitability of prejudice approach** *(alternative)*: Claims that as long as cultural stereotypes exist, prejudice inevitably follows; nonprejudiced responses are viewed as impression-management rather than genuine inhibition.
- **Tripartite model of attitudes** *(alternative)*: Treats stereotypes as the cognitive component of prejudiced attitudes, linking stereotype knowledge directly to prejudice

without distinguishing controlled inhibition.

- **Allport's early stereotype formation theory** *(background)*: Posits that stereotypes are established early in childhood and become part of the knowledge base.
- **Katz's well-learned association account** *(background)*: Describes stereotypes as strongly entrenched associative networks formed through repeated activation.
- **Automatic stereotype activation literature (Dovidio, Evans & Tyler 1986; Smith & Branscombe 1985)** *(background)*: Empirical work showing that well-learned racial stereotypes can be activated without conscious awareness.
- **Attentionless vs. detectionless processing (Bargh & Pietromonaco 1982; Greenwald, Klinger & Liu in press)** *(background)*: Distinguishes two routes to automatic activation that bypass conscious monitoring.
- **Functional view of prejudice (Billig 1985; LaViolette & Silvert 1951)** *(background)*: Frames prejudice as serving psychological functions rather than being a mere by-product of automatic activation.
- **Information-processing frameworks (Posner & Snyder 1975; Schneider & Shiffrin 1977; Logan & Cowan 1984)** *(background)*: General models of automatic and controlled processing that inform the dissociation model's mechanistic claims.

Study Design

- **Number of experiments:** 3
- **Predictors examined:** prejudice level (high vs. low), task condition (judgment vs. recognition vs. guess) - Experiment 2, priming proportion (80% vs. 20%) - within-subjects in Experiment 2, stimulus replication (Replication 1 vs. Replication 2) - within-subjects in Experiment 2
- **Outcomes examined:** Experiment 1: proportion of thoughts listed in each coding category (stereotype-knowledge measure), Experiment 2: vigilance reaction time, vigilance accuracy, impression-rating scores on hostility-related and unrelated scales, recognition hit rates (d'), guess response accuracy, and task-specific performance (judgment, recognition, guess), Experiment 3: proportion of pejorative alternate labels generated, counts of positive vs. negative thoughts, counts of trait vs. belief thoughts, presence of hostility themes in thought-listing protocols
- **Materials:** Paper booklets for free-response stereotype content (Exp 1); tachistoscopic word primes (Black-related and neutral) presented parafoveally with masks, plus the "Donald" paragraph and trait rating scales (Exp 2); in-house booklets for label-generation and thought-listing tasks together with the Modern Racism Scale (Exp 3).

- **Design types:**
- exp_1: between-subjects 2 (prejudice level)
- exp_2: mixed design - 3 (task condition) between-subjects × 2 (priming proportion) within-subjects × 2 (stimulus replication) within-subjects
- exp_3: between-subjects 2 (prejudice level)

Findings Summary

Across the three studies, high- and low-prejudice participants showed equivalent knowledge of the Black cultural stereotype (Study 1) and experienced similarly strong automatic activation of that stereotype, as evidenced by heightened hostility-related ratings after attentionless priming (Study 2). However, only low-prejudice individuals demonstrated controlled inhibition of the automatically activated stereotype: they generated more positive, belief-oriented thoughts and fewer pejorative labels, whereas high-prejudice participants produced more negative, trait-focused, and hostility-themed responses (Study 3). Thus, while automatic processes operate uniformly across prejudice levels, the capacity for controlled, nonprejudiced responses appears limited to low-prejudice subjects.

Evidence-Claim Alignment

Supported Claims

- High- and low-prejudice individuals are equally knowledgeable of the cultural Black stereotype (supported by Experiment 1 finding of no difference in stereotype content and judges' inability to predict prejudice level).
- Automatic activation of the Black stereotype occurs for both high- and low-prejudice participants (supported by Experiment 2 priming effect on hostility-related scales, with a significant Priming × Scale interaction showing stronger hostility ratings after 80% priming).
- Controlled inhibition of automatically activated stereotypes is possible for low-prejudice individuals (supported by Experiment 3 finding that low-prejudice subjects generated more positive, belief-oriented thoughts and fewer pejorative labels, indicating intentional suppression of stereotype-congruent content).

Unsupported or Weakly Supported Claims

- Controlled processes can inhibit automatic stereotype effects for all participants (the authors' general claim in the discussion is weakly supported because Experiment 3 showed inhibition only in low-prejudice subjects; high-prejudice participants did not differ in pejorative label generation and produced more negative, trait-focused thoughts).

- Automatic activation leads to a global negative evaluation beyond hostility (the prediction in Experiment 2's introduction is not fully supported; results reported only a Priming × Scale interaction for hostility-related scales, with no data on broader evaluative dimensions).

Contradictions or Tensions

- Experiment 1's outcome measure (proportion of thoughts in coding categories) does not directly assess stereotype knowledge, yet the authors claim it validates equal knowledge across prejudice levels (design-theory misalignment noted in the critique).
- The paper's discussion presents Study 3 as evidence that controlled processes can inhibit automatic effects, but the evidence is limited to low-prejudice participants, creating a tension between the claim of a general controlled-inhibition capability and the empirical pattern.
- Experiment 2's design includes a three-level between-subjects task condition, but the results section omits any main effects or interactions involving this factor, leaving a gap between the stated methodology and reported evidence.

Strengths

- Clear theoretical framing of automatic versus controlled processes, distinguishing stereotype knowledge from personal beliefs and linking them to prejudice levels.
- Use of three complementary experiments to test distinct model components: equal stereotype knowledge (Exp 1), automatic activation effects on ambiguous behavior (Exp 2), and controlled inhibition of stereotype-congruent thoughts (Exp 3).
- Consistent finding across Exp 1 and Exp 2 that high- and low-prejudice participants exhibit comparable stereotype knowledge and automatic activation, supporting the model's core claim of equal automatic processes.
- Experiment 3 provides converging evidence that low-prejudice individuals can intentionally suppress stereotype-congruent thoughts, demonstrating a controlled inhibition process.
- Inclusion of both explicit (Modern Racism Scale) and implicit (attentionless priming, thought-listing) measures to capture prejudice and stereotype activation from multiple methodological angles.
- Transparent reporting of participant numbers, stimulus materials, and coding categories for thought-listing, allowing replication of the core procedures.

Weaknesses and Concerns

- Outcome measure in Experiment 1 (proportion of coded thoughts) does not directly assess stereotype knowledge, creating a misalignment between the theoretical claim of equal knowledge and the empirical test.
- Authors overgeneralize the controlled-inhibition finding: Experiment 3 shows inhibition only for low-prejudice participants, yet the discussion presents it as a general capability of controlled processes.
- Experiment 2's introduction predicts a global negative evaluation beyond hostility, but the results report only a priming effect on hostility-related scales, leaving the broader prediction untested.
- Statistical reporting across experiments lacks effect sizes, confidence intervals, and corrections for multiple comparisons, limiting assessment of the practical significance and robustness of reported effects.
- Key design elements are insufficiently documented: no power analyses, absent counterbalancing procedures, and incomplete reporting of the between-subjects task condition in Experiment 2.

Key References

- **Allport (1954)** -- foundational theory
- **Katz (1976)** -- foundational theory
- **Dovidio, Evans, & Tyler (1986)** -- primary methodological precedent
- **Smith & Branscombe (1985)** -- primary methodological precedent
- **Bargh & Pietromonaco (1982)** -- primary methodological precedent
- **Logan & Cowan (1984)** -- primary methodological precedent
- **Neely (1977)** -- primary methodological precedent
- **Duncan (1976)** -- key empirical precedent
- **Nisbett & Wilson (1977)** -- key empirical precedent
- **Harding, Proshansky, Kutner, & Chein (1969)** -- competing theoretical account
- **Crosby, Bromley, & Saxe (1980)** -- competing theoretical account
- **Ehrlich (1973)** -- prior evidence for the same effect

Abstract

Preserved from source without summarization.

Three studies tested basic assumptions derived from a theoretical model based on the dissociation of automatic and controlled processes involved in prejudice. Study 1 supported the model's assumption that high- and low-prejudice persons are equally knowledgeable of the cultural stereotype. The model suggests that the stereotype is automatically activated in the presence of a member (or some symbolic equivalent) of the stereotyped group and that low-prejudice responses require controlled inhibition of the automatically activated stereotype. Study 2, which examined the effects of automatic stereotype activation on the evaluation of ambiguous stereotype-relevant behaviors performed by a race-unspecified person, suggested that when subjects' ability to consciously monitor stereotype activation is precluded, both high- and low-prejudice subjects produce stereotype-congruent evaluations of ambiguous behaviors. Study 3 examined high- and low-prejudice subjects' responses in a consciously directed thought-listing task. Consistent with the model, only low-prejudice subjects inhibited the automatically activated stereotype-congruent thoughts and replaced them with thoughts reflecting equality and negations of the stereotype. The relation between stereotypes and prejudice and implications for prejudice reduction are discussed.

Introduction (Paper-Level)

Section: paper-level

Research Gap / Motivation

It remains unclear how high- and low-prejudice individuals differ or are similar in their automatic and controlled responses to target group members, as only one prior study compared them.

Theoretical Framework

automatic vs controlled processes

> The present model ... suggests that automatic and controlled processes involved in stereotypes and prejudice can be dissociated, posits that the inevitability of prejudice arguments follow from tasks that are likely to engage automatic processes on which those high and low in prejudice are presumed not to differ.

stereotype vs personal belief structures

> Although they may have some overlapping features, it is argued that stereotypes and personal beliefs are conceptually distinct cognitive structures. Each structure represents part of one's entire knowledge base of a particular group.

Specific Hypotheses / Predictions

equal stereotype knowledge across prejudice levels

> The present model assumes that ... high- and low-prejudice persons are equally knowledgeable of the cultural stereotype of Blacks.

- Citations: Brigham (1972); Ehrlich (1973); Katz (1976); Proshansky (1966)

low-prejudice require controlled inhibition

> Because the stereotype has a longer history of activation ... overt nonprejudiced responses require intentional inhibition of the automatically activated stereotype and activation of the newer personal belief structure.

automatic activation influences ambiguous behavior evaluation

> Study 2 explored the implications of automatic racial stereotype priming on the evaluation of ambiguous stereotype-relevant behaviors.

Mechanistic Claims

automatic stereotype activation

> The stereotype ... is automatically activated in the presence of a member (or symbolic equivalent) of the target group.

- Citations: Dovidio, Evans, & Tyler (1986); Smith & Branscombe (1985)

controlled inhibition of stereotype

> Overt nonprejudiced responses require intentional inhibition of the automatically activated stereotype and activation of the newer personal belief structure.

conflict between stereotype and personal beliefs

> When an individual rejects the stereotype, he or she experiences a fundamental conflict between the already established stereotype and the more recently established personal beliefs.

Constructs of Interest

- **prejudice level (high vs low)** *(predictor)*: Individual differences in endorsement of negative stereotypes about Blacks.
- **responses to stereotyped group members** *(outcome)*: Evaluations, judgments, or behaviors that may reflect automatic stereotype activation or controlled inhibition.

Rationale for Predictor

Prejudice level is expected to determine the extent to which personal beliefs overlap with cultural stereotypes, influencing the need for controlled inhibition.

Rationale for Outcome

Responses reveal whether automatic stereotype activation or controlled processes dominate, thereby testing the model's dissociation of these processes.

Key Terminology

personal beliefs: Decisions about the appropriateness of stereotypic ascriptions; newer cognitive structures compared with long-standing stereotypes.

> personal beliefs (i.e., decisions about the appropriateness of stereotypic ascriptions) are necessarily newer cognitive structures

automatic processes: Unintentional or spontaneous activation of well-learned associations that occur without conscious effort and are inescapable.

> Automatic processes involve the unintentional or spontaneous activation of some well-learned set of associations or responses that have been developed through repeated activation in memory.

controlled processes: Intentional, effortful processes that require active attention and are flexible for decision making and behavior initiation.

> Controlled processes are intentional and require the active attention of the individual.

Background Research

- Allport (1954) showed that stereotypes are established early in children's memories.
- Katz (1976) documented that stereotypes are well-learned associations.
- Higgins & King (1981) demonstrated that gender stereotypes are automatically activated when the relevant category is not salient.
- Neely (1977) showed that automatic semantic priming can be overridden by conscious expectancies given sufficient time.
- Duncan (1976) found that ambiguous behavior is judged more negatively when performed by a Black actor, illustrating automatic stereotype priming.

Alternative Theories Discussed

inevitability of prejudice approach

- **Description:** Posits that as long as cultural stereotypes exist, prejudice inevitably follows because stereotypes are automatically applied to group members.
- **Difference from focal theory:** Emphasizes that knowledge of a stereotype is equated with prejudice, overlooking the distinction between stereotype knowledge and personal belief endorsement.
- **Citations:** Allport (1954); Ehrlich (1973)

tripartite model of attitudes

- **Description:** Treats stereotypes as the cognitive component of prejudiced attitudes.
- **Difference from focal theory:** Links stereotypes directly to prejudice rather than distinguishing them from personal beliefs and controlled inhibition.
- **Citations:** Harding, Proshansky, Kutner, & Chein (1969); Secord & Backman (1974)

Overview of Studies

Study 1 tests equal stereotype knowledge across prejudice levels; Study 2 examines automatic stereotype priming on ambiguous behavior evaluation; Study 3 assesses controlled inhibition via a thought-listing task.

Missing or Unclear

- Explicitly labeled hypothesis statements are not provided.
- Specific measurement instruments for prejudice or response outcomes are not described in the introduction.
- Statistical analysis plans are absent.

All Citations Referenced

- **Allport (1954)** -- early stereotype formation
- **Katz (1976)** -- well-learned associations
- **Porter (1971)** -- children's memory of stereotypes
- **Proshansky (1966)** -- developmental sequence
- **Higgins & King (1981)** -- gender stereotype activation
- **Dovidio, Evans, & Tyler (1986)** -- automatic activation
- **Smith & Branscombe (1985)** -- automatic activation

- **Brigham (1972)** -- socialization experiences
- **Ehrlich (1973)** -- socialization experiences
- **Greeley & Sheatsley (1971)** -- prejudice differences
- **Taylor, Sheatsley, & Greeley (1978)** -- prejudice differences
- **Duncan (1976)** -- automatic priming effects
- **Sager & Schofield (1980)** -- replication with children
- **Gaertner & McLaughlin (1983)** -- prejudice assessment
- **Harding, Proshansky, Kutner, & Chein (1969)** -- tripartite model
- **Secord & Backman (1974)** -- tripartite model
- **Billig (1985)** -- functional view of prejudice
- **LaViolette & Silvert (1951)** -- functional view
- **Tajfel (1981)** -- categorization and prejudice
- **Neely (1977)** -- automatic vs controlled processing
- **Logan (1980)** -- independence of processes
- **Logan & Cowan (1984)** -- independence of processes
- **Posner & Snyder (1975)** -- information-processing framework
- **Schneider & Shiffrin (1977)** -- information-processing framework
- **Shiffrin & Dumais (1981)** -- automatic processes

Additional Quoted Evidence

> The present model assumes that ... high- and low-prejudice persons are equally knowledgeable of the cultural stereotype of Blacks.

> Because the stereotype has a longer history of activation ... overt nonprejudiced responses require intentional inhibition of the automatically activated stereotype and activation of the newer personal belief structure.

> The inevitability of prejudice approach ... overlooks an important distinction between knowledge of a cultural stereotype and acceptance or endorsement of the stereotype.

Experiment 1 -- Introduction

Section: exp_1

Rationale for This Experiment

To validate the present model by directly examining whether high- and low-prejudice subjects differ in their knowledge of the cultural stereotype content about Blacks.

Relation to Prior Experiments

This is the first experiment in the series, establishing baseline knowledge of stereotype content before later experiments manipulate other variables.

New Elements Introduced

- Use of a free-response task instead of the typical adjective checklist to assess stereotype content
- Assessment of stereotype knowledge in both high-prejudice and low-prejudice participants

Constraints Addressed

- Addresses the lack of prior documentation on whether high- and low-prejudice individuals are equally knowledgeable about cultural stereotypes
- Provides a more sensitive measure of stereotype content by eliminating cueing from checklist items

Missing or Unclear

- Specific predictions tested in this experiment are not explicitly stated in the introduction
- Any theoretical refinements beyond the general model are not detailed

Citations Referenced

- **Brigham (1971)** -- implicit in the stereotype assessment literature
- **Gilbert (1951)** -- typical adjective checklist assessment of stereotype content
- **Karlins et al. (1969)** -- typical adjective checklist assessment of stereotype content
- **Katz & Braly (1933)** -- typical adjective checklist assessment of stereotype content

Experiment 1 -- Methods

Section: exp_1

Participants

- **N:** 40
- **N per condition:** high-prejudice: 21; low-prejudice: 19
- **Demographics:** White introductory psychology students
- **Compensation:** course credit
- **Testing context:** tested in groups of 4-6 in a lab setting; subjects isolated from each other after instructions

Design

Design statement: 2 (prejudice level: high vs. low) between-subjects design.

Predictors:

- **prejudice level**
- Levels: high, low
- Manipulation: between-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **proportion of thoughts listed in each coding category:** percentage of listed stereotype components falling into each predefined coding category

Materials

- **Description:** paper booklet with blank lines for participants to list components of the cultural stereotype of Blacks; Modern Racism Scale questionnaire
- **Source:** in-house (created for the study) and Modern Racism Scale (McConahay, Hardee, & Batts, 1981)
- **Additional details:**
- coding categories defined by researchers

Procedure

Ordered steps:

1. General written instructions were given to participants.
2. Participants listed components of the Black stereotype on a blank-page booklet.
3. Participants completed the seven-item Modern Racism Scale.
4. Completed booklets were placed in unmarked envelopes and deposited in a box.
5. Participants were debriefed and thanked.

- **Apparatus:** paper booklets, envelopes, collection box; no computer equipment reported
- **Experimenter presence:** experimenter left the room after giving general instructions
- **Presentation format:** not reported
- **Stimulus timing:** not reported

Verbatim instructions (as quoted in the paper):

> the questionnaire was designed to help researchers better understand social stereotypes and that interest centered on the cultural stereotype of Blacks.

> she was not interested in their personal beliefs but in their knowledge of the content of the cultural stereotype.

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Modern Racism Scale is designed to measure anti-Black attitudes in a nonreactive fashion

Missing or Unclear

- recruitment source not specified
- detailed demographics beyond race not provided
- exclusion criteria not reported
- stimulus timing and presentation details not reported
- session duration not reported
- counterbalancing scheme not described
- apparatus details beyond paper materials not reported
- data analysis plan not described

- preregistration status not reported
- power analysis not reported

Citations Referenced in This Section

- **McConahay, Hardee, & Batts (1981)** -- source of the Modern Racism Scale used to assess anti-Black attitudes
- **Kinder & Sears (1981)** -- supporting the predictive validity of the Modern Racism Scale
- **Sears & Kinder (1971)** -- supporting the predictive validity of the Modern Racism Scale
- **Sears & McConahay (1973)** -- supporting the predictive validity of the Modern Racism Scale

Quoted Evidence from Source

> Forty White introductory psychology students participated in groups of 4-6 for course credit.

> Subjects were assigned to a high-prejudice (N = 21) or a low-prejudice (N = 19) group on the basis of a median split of scores on the scale.

> Written instructions told subjects that the questionnaire was designed to help researchers better understand social stereotypes and that interest centered on the cultural stereotype of Blacks.

Experiment 1 -- Results and Discussion

Section: exp_1

Statistical Approach

The prediction of no difference between the high- and low-prejudice subjects' knowledge of the cultural stereotype was tested in two different ways. First, none of the differences in Table 1 was statistically reliable. Second, two separate judges were given subjects' protocols and were instructed to read the content listed and to separate the protocols into high- and low-prejudice groups.

Key Findings

High- and low-prejudice subjects did not differ in their knowledge of the cultural stereotype.

- **Test:** Chi-square tests and judges' predictions
- **Effect type:** main effect
- **Variables in model:** Prejudice level, Stereotype content

Verbatim statistical report:

> none of the differences in Table 1 was statistically reliable. The judges could not reliably predict the subjects' prejudice level from the content of their protocols.

- Citations: Ehrlich (1973)

Authors' Interpretations

Aggressiveness in stereotypes

> the most common theme in subjects' protocols was that Blacks are aggressive, hostile, or criminal-like

- Citations: Donnerstein & Donnerstein (1972); Donnerstein, Donnerstein, Simon, & Ditrichs (1972); Duncan (1976); Sager & Schofield (1980)

Stereotype content similarity

> there appeared to be few differences in the content reported by high- and low-prejudice subjects

Authors' Stated Links to Their Own Predictions

No difference in stereotype knowledge

> These data validate Ehrlich's (1973) assumption as well as the first assumption of the present model: High- and low-prejudice persons are indeed equally knowledgeable of the cultural stereotype.

- Citations: Ehrlich (1973)

Missing or Unclear (Non-Statistical)

- Table 1 contents not provided in the text

Citations Referenced

- **Donnerstein & Donnerstein (1972)** -- research on the effect of racial stereotypes on perception and behavior

- **Donnerstein, Donnerstein, Simon, & Ditrichs (1972)** -- research on the effect of racial stereotypes on perception and behavior
- **Duncan (1976)** -- research on the effect of racial stereotypes on perception and behavior
- **Sager & Schofield (1980)** -- research on the effect of racial stereotypes on perception and behavior
- **Ehrlich (1973)** -- assumption about stereotype knowledge

Experiment 2 -- Introduction

Section: exp_2

Rationale for This Experiment

Study 1 used a controlled free-response task, which cannot reveal implicit cognitive structures. This experiment was designed to use a nonconscious priming task that eliminates controlled processes, allowing examination of automatic stereotype activation effects in high- and low-prejudice participants.

Relation to Prior Experiments

Experiment 2 builds on Study 1 by replacing the controlled free-response measure with an implicit, nonconscious priming paradigm to test whether automatic processes alone influence judgments of ambiguous behavior.

Specific Predictions

automatic activation increases hostility ratings

> Because hostility is part of the racial stereotype, increased priming should lead to more extreme ratings on the hostility-related scales for both high- and low-prejudice subjects.

global negative evaluation beyond hostility

> The remaining trait scales were not related to hostility and provided the opportunity to examine the possibility that stereotype activation led...a global negative evaluation that generalized beyond hostility ratings.

Theoretical Refinements (relative to paper-level intro)

attentionless processing as automatic activation

> In this study attentionless processing was accomplished by presenting the primes parafoveally (Bargh & Pietromonaco,1982) followed immediately with a pattern mask.

- Citations: Bargh & Pietromonaco (1982)

detectionless vs attentionless automatic processing

> Greenwald, Klinger, and Liu (in press) recently suggested that automatic activation can be achieved through either detectionless processing or attentionless processing, both of which have been shown to produce reliable priming effects.

- Citations: Greenwald, Klinger, and Liu (in press)

New Elements Introduced

- Nonconscious priming via parafoveal word presentation (attentionless processing)
- Vigilance task with 20% vs 80% stereotype-related primes
- Impression-formation task with ambiguous hostile behavior description
- Trait rating scales separating hostility-related and unrelated dimensions

Constraints Addressed

- Eliminates controlled response confounds present in Study 1
- Tests automatic activation criteria debated in the literature
- Provides a dissociation between automatic and controlled processes

Missing or Unclear

- Explicit prediction about differential effects for high vs. low prejudice (whether magnitude differs) is not clearly stated

Citations Referenced

- **Carver, Ganellin, Froming, & Chambers (1983)** --
- **Srull & Wyer (1979,1980)** --
- **Bargh, Bond, Lombardi, & Tota (1986)** --
- **Bargh & Pietromonaco (1982)** --
- **Brigham (1971)** --
- **Duncan (1976)** --

- **Sager & Schofield (1980)** --
- **Stephan (1985)** --
- **Holender (1986)** --
- **Marcel (1983b)** --
- **Greenwald, Klinger, and Liu (in press)** --
- **Bolota (1983)** --
- **Fowler, Wolford, Slade, & Tassinary (1981)** --
- **Klatzky (1984)** --

Experiment 2 -- Methods

Section: exp_2

Participants

- **N:** 122
- **Demographics:** White subjects; age, gender, education not reported
- **Recruitment:** Introductory psychology students at the university
- **Compensation:** Extra credit
- **Testing context:** Individual lab sessions; vigilance task 11-13 min plus additional tasks
- **Exclusions:**
- 3 Black subjects replaced
- 1 subject with dyslexia
- 3 subjects who failed to follow instructions

Design

Design statement: 3 (task condition: judgment vs recognition vs guess) between-subjects design, with stimulus priming proportion (80% vs 20%) and replication (1 vs 2) varied within each condition.

Predictors:

- **task condition**
- Levels: judgment, recognition, guess
- Manipulation: between-subjects

- Factor type: nominal
- **priming proportion**
- Levels: 80%, 20%
- Manipulation: within-subjects
- Factor type: nominal
- **stimulus replication**
- Levels: Replication 1, Replication 2
- Manipulation: within-subjects
- Factor type: nominal

Counterbalancing: not reported

Outcome Measures

- **vigilance reaction time:** latency recorded by Hunter Model 120 clock to nearest millisecond for left/right response
- **vigilance accuracy:** correct identification of stimulus side (left/right)
- **impression rating scores:** 12 trait scales rated 0-10 after reading the Donald paragraph
- **recognition hits:** number of correctly identified previously presented words on the 48-item recognition test
- **guess responses:** participants' reported guesses for each word presented in the guess condition

Materials

- **Description:** Black and neutral words used as primes; stimuli presented tachistoscopically with mask
- **Source:** Adapted from Bargh & Pietromonaco (1982); neutral words from Carroll, Davies, & Richman (1971)
- **Stimulus pool size:** 100 words per replication (80% and 20% lists)
- **Stimuli presented per participant:** 100 experimental trials per participant (plus 10 practice trials)
- **Additional details:**
- Each word presented for 80 ms
- Interstimulus interval 2-7 s

- Words centered in quadrants, distance 2.3 in from fixation

Procedure

Ordered steps:

1. Screen Intro psychology students with Modern Racism Scale
2. Select upper and lower third scorers
3. Phone screening for perfect vision
4. Replace ineligible subjects
5. Conduct vigilance task (10 practice, 100 experimental trials)
6. Proceed to condition-specific second task (judgment, recognition, or guess)
7. Debrief participants

- **Apparatus:** Scientific Prototype two-channel tachistoscope, experimenter-controlled panel, Hunter Model 120 clock, two-button response box
- **Experimenter presence:** Experimenter present in the room; blind to prejudice level, priming condition, and replication condition
- **Presentation format:** Randomized order within blocks; 25 of 100 trials per replication assigned to each quadrant
- **Stimulus timing:** 80 ms stimulus, immediate mask, ISI 2-7 s
- **Session duration:** Vigilance task 11-13 min; total session length not explicitly reported

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Method modeled after Bargh & Pietromonaco (1982)
- Tachistoscopic presentation used instead of computer monitor

Missing or Unclear

- Age, gender, and education demographics not reported
- Power analysis or sample size justification not provided
- Randomization or counterbalancing details for condition assignment not described

- Exact debrief script not provided
- Compensation details beyond extra credit not specified
- Full session duration including second task not reported

Citations Referenced in This Section

- **Bargh and Pietromonaco (1982)** -- Method and procedure modeled after this study
- **Srull and Wyer (1979, 1980)** -- Source of the "Donald" paragraph used in judgment condition
- **Carroll, Davies, & Richman (1971)** -- Source of high-frequency neutral words

Quoted Evidence from Source

> After replacing 3 Black subjects,1 subject who reported having dyslexia following the vigilance task, and 3 subjects who failed to follow instructions, the sample consisted of 78 White subjects in the judgment condition,32 White subjects in the recognition condition, and 12 White subjects in the guess condition.

> The experimenter remained blind to subjects' prejudice level, priming condition, and stimulus replication condition.

> Each stimulus was presented for 80 ms and was immediately followed by a mask (a jumbled series of letters).

Experiment 2 -- Results and Discussion

Section: exp_2

Statistical Approach

The primary analysis concerned whether subjects performed the recognition task better than would be expected by chance. The hits and false alarms were used to generate d' scores for both stereotype-related and neutral words. The d' scores were submitted to a four-way mixed-model analysis of variance (ANOVA)--Prejudice Level × Priming × Replication × Word Type--with word type (stereotype-related vs. neutral) as a repeated measure. The mean ratings were submitted to a mixed-model ANOVA, with prejudice level (high vs. low), priming (20% vs. 80%), and replication (1 vs. 2) as between-subjects variables and scale (hostility related vs. hostility unrelated) as a within-subjects variable.

Key Findings

Subjects guessed few words accurately in the guess condition

- **Test:** descriptive statistics
- **Effect type:** descriptive statistics
- **Variables in model:** word type (stereotype-related vs. neutral), prejudice level (high vs. low)

Verbatim statistical report:

> Of the 1,200 guesses, subjects guessed 20 words accurately, a hit rate of 1.67%. Overall, subjects guessed 1.4% of the stereotype-related words and 3.33% of the neutral words.

- Citations: Bargh and Pietromonaco (1982)

Subjects could not reliably recognize primes in the recognition condition

- **Test:** d' scores and t-tests
- **Effect type:** main effect
- **Variables in model:** word type (stereotype-related vs. neutral), prejudice level (high vs. low)

Verbatim statistical report:

> Over all subjects, neither d' for stereotype-related words ($M = .01$) nor for neutral words ($M = .07$) differed significantly from zero ($ps > .42$). High-prejudice subjects' mean d' scores for stereotyped-related and neutral words were .02 and .12, respectively ($ps > .40$). Low-prejudice subjects' mean d' scores for stereotype-related and neutral words were .01 and .02, respectively ($p > .84$).

- Citations: Green and Swets (1966)

Prejudice level did not affect recognition performance

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** prejudice level, priming, replication, word type

Verbatim statistical report:

> Prejudice level did not affect subjects' overall performance, $F(1,24) = 0.07$, $p = .78$, and it did not interact with word type, $F(1,24) = 0.04$, $p = .84$.

Priming did not interact with word type or prejudice level in recognition performance

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** prejudice level, priming, replication, word type

Verbatim statistical report:

> Priming did not interact with word type, $F(1,24) = 0.47$, $p = .50$, or affect the Prejudice × Word Type interaction, $F(1,24) = 0.32$, $p < .56$.

Priming affected hostility-related ratings but not hostility-unrelated ratings

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** prejudice level, priming, replication, scale (hostility-related vs. hostility-unrelated)

Verbatim statistical report:

> The Priming × Scale interaction was significant, $F(1,70) = 5.04$, $p < .03$. Ratings on the hostility-related scales were more extreme in the 80% ($M = 7.52$) than in the 20% ($M = 6.87$) priming condition. The hostility-unrelated scales were unaffected by priming ($Ms = 5.89$ and 6.00 for the 20% and 80% priming conditions, respectively).

- Citations: Srull and Wyer (1979); Bargh and Pietromonaco (1982)

Prejudice level did not interact with priming on hostility-related scales

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** prejudice level, priming, replication, scale (hostility-related vs. hostility-unrelated)

Verbatim statistical report:

> The Prejudice Level × Priming interaction was nonsignificant, $F(1,70) = 1.19$, $p = .28$. The analysis on hostility-related scales revealed only a significant priming main effect, $F(1,70) = 7.59$, $p < .008$.

Priming did not affect hostility-unrelated scales

- **Test:** ANOVA
- **Effect type:** main effect
- **Variables in model:** prejudice level, priming, replication, scale (hostility-related vs. hostility-unrelated)

Verbatim statistical report:

> In the analysis of the hostility-unrelated scales, neither the priming main effect, $F(1,70) = 0.23$, $p = .63$, nor the Prejudice Level × Priming interaction, $F(1,70) = 0.02$, $p = .88$, reached significance.

Prejudice level interacted with priming and replication

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** prejudice level, priming, replication, scale (hostility-related vs. hostility-unrelated)

Verbatim statistical report:

> The Prejudice Level × Priming × Replication interaction, $F(1,70) = 4.69$, $p < .03$, indicated that the priming effect was slightly reversed for low-prejudice subjects exposed to Replication 1.

Prejudice level interacted with scale relatedness and replication

- **Test:** ANOVA
- **Effect type:** interaction
- **Variables in model:** prejudice level, priming, replication, scale (hostility-related vs. hostility-unrelated)

Verbatim statistical report:

> A Prejudice Level × Scale Relatedness × Replication interaction, $F(1,70) = 4.42$, $p < .04$, suggested that the difference between scores on hostility-related and hostility-unrelated scales was greater for low-prejudice subjects in Replication 1 and high-prejudice subjects in Replication 2.

Authors' Interpretations

attentionless processing

> These data suggest that neither high- nor low-prejudice subjects were able to identify the content of the priming words at the point of encoding, thus satisfying one criterion for attentionless processing.

conscious access to primes

> Subjects were not able to reliably recognize either stereotype-related or neutral words, suggesting that subjects did not have conscious access to the content of the primes, thus establishing the second criterion for attentionless processing.

automatic stereotype activation

> These findings extend those of Srull and Wyer (1979,1980), Bargh and Pietromonaco (1982), Bargh et al. (1986), and Carver et al. (1983) in demonstrating that in addition to trait categories, stereotypes can be primed and can affect the interpretation of subsequently encoded social information.

- Citations: Srull and Wyer (1979,1980); Bargh and Pietromonaco (1982); Bargh et al. (1986); Carver et al. (1983)

stereotype activation and hostility

> The data are consistent with Duncan's (1976) hypothesis that priming the racial stereotype activates a link between Blacks and hostility.

- Citations: Duncan (1976)

stereotypes and prejudiced responses

> These data, however, should not be interpreted as suggesting that all people are prejudiced. It could be argued that neither task allowed for the possibility of nonprejudiced responses.

automatic stereotype activation and prejudice

> The present data suggest that when automatically accessed the stereotype may have effects that are inaccessible to the subject (Nisbett & Wilson,1977). Thus, even for subjects who honestly report having no negative prejudices against Blacks, activation of stereotypes can have automatic effects that if not consciously monitored produce effects that resemble prejudiced responses.

- Citations: Nisbett & Wilson (1977)

Authors' Stated Links to Their Own Predictions

priming effect on hostility-related scales

> These results were consistent with the present model and suggest that the effects of automatic stereotype priming were equally strong for high- and low-prejudice subjects.

stereotype activation and hostility

> The data are consistent with Duncan's (1976) hypothesis that priming the racial stereotype activates a link between Blacks and hostility.

- Citations: Duncan (1976)

Limitations Stated by Authors

- It could be argued that neither task allowed for the possibility of nonprejudiced responses. Study 1 encouraged subjects not to inhibit prejudiced responses. Study 2 suggested that when the racial category is activated and subjects' ability to consciously monitor this activation is bypassed, their responses reflect the activation of cognitive structures with a longer history (i.e., greater frequency) of activation.

Preview of Next Experiment

Study 3 examined the responses of high- and low-prejudice subjects to a task designed to focus attention on and thus activate subjects' personal beliefs about Blacks (in addition to the automatically activated stereotype).

Missing or Unclear (Non-Statistical)

- No specific descriptive statistics (e.g., means, standard deviations) were provided for the incorrect guesses related to the racial stereotype.
- No specific descriptive statistics (e.g., means, standard deviations) were provided for the higher-order interactions involving prejudice level, priming, replication, and scale relatedness.

Citations Referenced

- **Bargh and Pietromonaco (1982)** -- replicating their findings
- **Green and Swets (1966)** -- tabled d' scores
- **Srull and Wyer (1979,1980)** -- extending their findings
- **Bargh and Pietromonaco (1982)** -- extending their findings
- **Bargh et al. (1986)** -- extending their findings
- **Carver et al. (1983)** -- extending their findings

- **Duncan (1976)** -- consistent with his hypothesis
- **Nisbett & Wilson (1977)** -- supporting the idea of inaccessible effects
- **Higgins and King (1981)** -- culturally defined stereotypes
- **Bettleheim and Janowitz (1964)** -- knowledge of stereotypes

Experiment 3 -- Introduction

Section: exp_3

Rationale for This Experiment

To test whether making the stereotype-personal belief conflict salient causes low-prejudice individuals to engage controlled inhibition of stereotypic thoughts, contrasting their responses with high-prejudice individuals.

Relation to Prior Experiments

Builds on earlier studies by directly manipulating conflict salience (anonymous thought-listing) to examine controlled inhibition processes hypothesized in the model.

Specific Predictions

low-prejudice thought content

> Low-prejudice subjects ... will take this opportunity to demonstrate that they do not endorse the cultural stereotype; they are likely to inhibit stereotype-congruent thoughts and intentionally replace them with thoughts consistent with their nonprejudiced personal beliefs.

high-prejudice thought content

> High-prejudice subjects, because their beliefs overlap with the stereotype, are expected to list stereotype-congruent thoughts.

controlled inhibition requirement

> Resolution of the conflict ... requires controlled inhibition (Logan & Cowan,1984; Neely,1977) of the automatically activated stereotype.

- Citations: Logan & Cowan (1984); Neely (1977)

Theoretical Refinements (relative to paper-level intro)

identity threat motivates inhibition

> When their nonprejudiced identity is threatened, low-prejudice persons are motivated to reaffirm their nonprejudiced self-concepts (Dutton,1976; Dutton & Lake,1973).

- Citations: Dutton (1976); Dutton & Lake (1973)

conflict salience triggers controlled processes

> When the conflict between their nonprejudiced personal beliefs and the stereotype of Blacks is made salient, low-prejudice persons are likely to resolve the conflict by denouncing the stereotype and expressing their nonprejudiced beliefs.

New Elements Introduced

- Anonymous thought-listing task about Blacks to make stereotype-belief conflict salient
- Comparison of high- vs low-prejudice participants on content of written thoughts

Constraints Addressed

- Addresses how identity threat influences the effort put into stereotype inhibition
- Tests the role of controlled inhibition in resolving stereotype-belief conflict

Citations Referenced

- **Dutton (1976)** --
- **Dutton & Lake (1973)** --
- **Logan & Cowan (1984)** --
- **Neely (1977)** --

Experiment 3 -- Methods

Section: exp_3

Participants

- **N:** 67 White introductory psychology students
- **Demographics:** White, introductory psychology students, received course credit
- **Recruitment:** introductory psychology course
- **Compensation:** course credit

- **Testing context:** Groups of 3-6 seated at partitioned tables in a lab setting

Design

Design statement: not reported

Counterbalancing: not reported

Outcome Measures

- **label generation:** number of alternate labels listed within 1 minute
- **thought listing:** number of thoughts recorded in 10 boxes within 10 minutes
- **Modern Racism Scale:** score on the seven-item Modern Racism Scale

Materials

- **Description:** Booklet containing label-generation task, thought-listing boxes, and a seven-item Modern Racism Scale
- **Source:** in-house materials
- **Additional details:**
- 1-min label-generation period
- 10-min thought-listing period
- two pages of 10 thought-listing boxes

Procedure

Ordered steps:

1. Collect and sign participation cards; store them anonymously
2. Read general instructions on first page of booklet
3. Label-generation task: list alternate labels for Black Americans (1 min)
4. Read thought-listing instructions
5. Thought-listing task: record thoughts about Black Americans and generated labels (10 min)
6. Complete the seven-item Modern Racism Scale
7. Read debriefing document

- **Apparatus:** paper booklets, pens, participation cards

- **Experimenter presence:** present
- **Presentation format:** not reported
- **Stimulus timing:** 1 min for label task; 10 min for thought-listing task

Preregistration and Analysis Plan

- **Preregistration:** not reported

Author-Provided Rationale for Methodological Choices

- Anonymity procedures (partitioned tables, collection of signed cards) were used to enhance anonymity and reduce inhibition

Missing or Unclear

- explicit experimental design (e.g., between-subjects, within-subjects) not described
- predictor variables and their levels not specified
- counterbalancing scheme not reported
- stimulus details (e.g., specific labels or thought prompts) not provided
- exact timing parameters beyond overall limits not given (e.g., ISI)
- operational definitions of outcome measures beyond general description not detailed
- session total duration not reported
- power analysis not reported
- preregistration status not reported
- data analysis plan not reported

Quoted Evidence from Source

> Subjects were 67 White introductory psychology students who participated for course credit.

> Subjects were run in groups of 3-6 and were seated at partitioned tables so that subjects were isolated from each other.

> They were allowed 1 min to complete this task.

Experiment 3 -- Results and Discussion

Section: exp_3

Statistical Approach

The frequency of these positive-belief, negative-belief, and trait thoughts listed in subjects' protocols were submitted to a Prejudice Level (high vs. low) × Valence (positive vs. negative) × Thought Type (trait vs. belief) mixed-model ANOVA. Prejudice level was a between-subjects variable, and valence and thought type were within-subjects variables.

Key Findings

High- and low-prejudice subjects did not differ in the proportion of pejorative alternate labels generated

- **Test:** independent samples t-test
- **Effect type:** main effect
- **Variables in model:** prejudice level, proportion of pejorative labels

Verbatim statistical report:

> $t(64) = .68$, $p > .10$

Verbatim descriptives:

> High-prejudice ($M = .53$) and low-prejudice ($M = .44$) subjects

High-prejudice subjects listed more negative than positive thoughts, and low-prejudice subjects listed more positive than negative thoughts

- **Test:** mixed-model ANOVA
- **Effect type:** two-way interaction
- **Variables in model:** prejudice level, valence, thought type

Verbatim statistical report:

> $F(1,65) = 28.82$, $p < .0001$

Verbatim descriptives:

> High-prejudice: negative ($M = 2.06$) vs. positive ($M = 1.48$); Low-prejudice: positive ($M = 2.28$) vs. negative ($M = 1.10$)

- Citations: Brigham (1971)

High-prejudice subjects were more likely to list trait than belief thoughts, while low-prejudice subjects were more likely to list belief than trait thoughts

- **Test:** mixed-model ANOVA
- **Effect type:** two-way interaction
- **Variables in model:** prejudice level, valence, thought type

Verbatim statistical report:

> $F(1,65) = 18.04$, $p < .0001$

Verbatim descriptives:

> High-prejudice: trait ($M = 2.56$) vs. belief ($M = 1.52$); Low-prejudice: belief ($M = 2.86$) vs. trait ($M = 1.12$)

- Citations: Brigham (1971)

High-prejudice subjects most often listed negative traits, and low-prejudice subjects most frequently listed positive-belief thoughts

- **Test:** mixed-model ANOVA with post hoc Duncan test
- **Effect type:** three-way interaction
- **Variables in model:** prejudice level, valence, thought type

Verbatim statistical report:

> $F(1,65) = 4.88$, $p < .03$; Duncan test ($p = .05$)

Verbatim descriptives:

> High-prejudice: negative trait ($M = 3.32$) vs. positive-belief ($M = 1.17$), negative-belief ($M = 1.18$), positive trait ($M = 1.79$); Low-prejudice: positive-belief ($M = 4.52$) vs. negative-belief ($M = 1.21$), positive trait ($M = 1.24$), negative trait ($M = 1.00$)

High-prejudice subjects were more likely to include hostility themes in their thought-listing protocols than low-prejudice subjects

- **Test:** z test on proportions
- **Effect type:** main effect
- **Variables in model:** prejudice level, presence of hostility themes

Verbatim statistical report:

> $z = 4.41$, $p < .01$

Verbatim descriptives:

> High-prejudice: 60%; Low-prejudice: 9%

Authors' Interpretations

High- and low-prejudice subjects' thoughts

> Taken together, these sets of analyses indicate that high- and low-prejudice subjects were willing to report different thoughts about Blacks. In addition, these analyses suggested that there were sufficient levels of variability in prejudice levels among the subjects to detect the effects of prejudice in the previous studies should those effects exist. The thought-listing task was one in which subjects were likely to think carefully about what their responses implied about their prejudice-relevant self-concepts. For those who valued a nonprejudiced identity, writing stereotype-congruent thoughts would have been inconsistent with and perhaps would have threatened their nonprejudiced identity.

- Citations: Neely (1977)

Low-prejudice subjects' thoughts

> Thus, even under anonymous conditions, low-prejudice subjects apparently censored and inhibited (Neely,1977) the automatically activated negative stereotype-congruent information and consciously replaced it with thoughts that expressed their nonprejudiced values. Low-prejudice subjects wrote few pejorative thoughts. Their thoughts were more likely to have reflected the importance of equality or the negation of the cultural stereotype. Moreover, low-prejudice subjects appeared reluctant to ascribe traits to the group as a whole.

- Citations: Neely (1977)

High-prejudice subjects' thoughts

> In contrast, the protocols of high-prejudice subjects seemed much more consistent with the cultural stereotype of Blacks. Their thoughts were primarily negative, and they seemed willing to ascribe traits to the group (especially negative traits).

Authors' Stated Links to Their Own Predictions

Prejudice Level × Valence interaction

> The analysis revealed the expected Prejudice Level × Valence interaction, $F(1,65) = 28.82$, $p < .0001$. High-prejudice subjects listed more negative ($M = 2.06$) than positive ($M = 1.48$) thoughts, and lowprejudice subjects listed more positive ($M = 2.28$) than negative ($M = 1.10$) thoughts.

Prejudice Level × Type interaction

> In addition, there was a Prejudice Level × Type interaction, $F(1,65) = 18.04$, $p < .0001$. This interaction suggested that high-prejudice subjects were more likely to list trait ($M = 2.56$) than belief ($M = 1.52$) thoughts. In contrast, low-prejudice subjects were more likely to list belief ($M = 2.86$) than trait ($M = 1.12$) thoughts.

- Citations: Brigham (1971)

Missing or Unclear (Non-Statistical)

- The coding scheme for thoughts generated is not fully described in the text.

Citations Referenced

- **Brigham (1971)** -- Black stereotype traditionally has been largely negative and composed of traits
- **Neely (1977)** -- censored and inhibited (Neely,1977) the automatically activated negative stereotype-congruent information and consciously replaced it with thoughts that expressed their nonprejudiced values
- **Donnerstein et al. (1972)** -- Much of the intergroup perception literature has assumed that the hostility component of the stereotype influences perceptions of Blacks (Donnerstein et al.,1972; Duncan,1976; Sager & Schofield,1980)
- **Duncan (1976)** -- Much of the intergroup perception literature has assumed that the hostility component of the stereotype influences perceptions of Blacks (Donnerstein et al.,1972; Duncan,1976; Sager & Schofield,1980)
- **Sager & Schofield (1980)** -- Much of the intergroup perception literature has assumed that the hostility component of the stereotype influences perceptions of Blacks (Donnerstein et al.,1972; Duncan,1976; Sager & Schofield,1980)

General Discussion

Section: paper-level

Summary of Findings

Studies 1 and 2 show that both high- and low-prejudice individuals possess stereotype knowledge and exhibit equally strong automatic activation. Study 3 demonstrates that controlled, intentional processes can inhibit these automatic effects when individuals are motivated to maintain a nonprejudiced identity.

Theoretical Conclusions

controlled inhibition of automatic stereotypes

> Nonprejudiced responses are, according to the dissociation model, a function of intentional, controlled processes and require a conscious decision to behave in a nonprejudiced fashion.

prejudice not inevitable

> Thus, in contrast to the pessimistic analysis by Crosby et al. (1980), the present framework suggests that rather than all people being prejudiced, all are victims of being limited capacity processors.

- Citations: Crosby et al. (1980)

habit-breaking analogy for stereotype inhibition

> Inhibiting stereotype-congruent or prejudice-like responses and intentionally replacing them with nonprejudiced responses can be likened to the breaking of a bad habit.

Cross-Experiment Integration

integration of studies 1-3

> Study 1 suggested both high- and low-prejudice persons possess knowledge of the racial stereotype... Study 2 suggested that automatic stereotype activation is equally strong... Study 3, however, provided evidence that controlled processes can inhibit the effects of automatic processing when the implications of such processing compete with goals to establish or maintain a nonprejudiced identity.

Results Authors State Support Their Theory

equal stereotype knowledge across prejudice levels

> Study 1 suggested both high- and low-prejudice persons possess knowledge of the racial stereotype.

equal automatic activation across prejudice levels

> Study 2 suggested that automatic stereotype activation is equally strong and equally inescapable for high- and low-prejudice subjects.

controlled processes can inhibit automatic effects

> Study 3, however, provided evidence that controlled processes can inhibit the effects of automatic processing when the implications of such processing compete with goals to establish or maintain a nonprejudiced identity.

Results Authors State Challenge a Theory

challenge to inevitability of prejudice

> Crosby, Bromley, and Saxe (1980) argued that the inconsistency ... is evidence that (all) White Americans are prejudiced ... I disagree fundamentally with this premise.

- Citations: Crosby et al. (1980)

Alternative Accounts Addressed by Authors

inevitability of prejudice (Crosby et al., 1980) *(named theory)*

- **Description:** All White Americans are prejudiced; nonprejudiced responses are impression-management attempts.
- **How authors addressed it:** argued against; claim the premise denies possibility of attitude change and is a limitation
- **Citations:** Crosby et al. (1980)

Implications for the Field

need for detailed model of controlled inhibition

> What is needed now is a fully articulated model of controlled processes that delineates the cognitive mechanisms involved in inhibition.

- Citations: Logan and Cowan (1984); Bargh (1984)

exploration of variables engaging controlled processes

> The present framework highlights the potential for nonprejudiced behaviors when social desirability concerns are minimal (Study 3) and invites researchers to explore the variables that are likely to engage controlled stereotype-inhibiting processes in

intergroup settings.

Future Directions Proposed by Authors

- Develop a fully articulated model of controlled processes that delineates the cognitive mechanisms involved in inhibition.
- Systematically explore the variables that engage controlled stereotype-inhibiting processes in intergroup settings.

Missing or Unclear

- Authors do not explicitly state limitations of their own work.
- No specific applied implications are articulated.

Citations Referenced

- **Pettigrew (1987)** --
- **Crosby et al. (1980)** --
- **Baxter (1973)** --
- **Gaertner (1976)** --
- **Gaertner & Dovidio (1977)** --
- **Linn (1965)** --
- **Weitz (1972)** --
- **Logan and Cowan (1984)** --
- **Bargh (1984)** --
- **Ronis, Yates, and Kirscht (in press)** --
- **Greeley & Sheatsley (1971)** --
- **Taylor et al. (1978)** --
- **Higgins & King (1981)** --

Critique

Paper: devine-1989-stereotypes-and-prejudice--their-automatic-and-controlled-components

Overall Assessment

The paper suffers from several moderate methodological and analytical shortcomings that undermine its scholarly rigor. The outcome measures and theoretical alignment are not tightly matched, and key statistical results lack necessary follow-up analyses, effect sizes, and corrections for multiple comparisons, limiting confidence in the reported effects. Additionally, insufficient reporting of power analyses, counterbalancing, and reliability estimates further weakens the robustness of the findings. While the study addresses an important topic, these issues collectively prevent the work from meeting high standards of methodological rigor and evidentiary support.

1. Design-Theory Alignment

- **In Experiment 1 the outcome measure (proportion of thoughts listed in coding categories) does not directly assess the intended construct of stereotype knowledge, which the study aims to test.**
- Severity: **moderate**
- Justification: The theoretical claim is that high- and low-prejudice individuals possess equal knowledge of the cultural stereotype. Measuring "proportion of thoughts" after a free-response task conflates recall with coding decisions and does not provide a clear, validated index of knowledge. This misalignment weakens the test of the hypothesis that knowledge is equal across prejudice levels.
- Evidence:
- `methods/exp_1` -- field `outcome_measures`: name: proportion of thoughts listed in each coding category; operationalization: percentage of listed stereotype components falling into each predefined coding category
- `introduction/exp_1` -- field `rationale_for_this_experiment`: to directly examine whether high- and low-prejudice subjects differ in their knowledge of the cultural stereotype content about Blacks

2. Evidence-Theory Alignment

- **The authors overstate that controlled processes can inhibit automatic stereotype effects for all participants, despite results showing only low-prejudice subjects exhibited inhibition while high-prejudice subjects did not.**
- Severity: **moderate**
- Justification: The theoretical claim that controlled inhibition can counteract automatic stereotype activation is central to the model. However, the empirical data from Study 3 demonstrate inhibition effects exclusively in low-prejudice participants, with high-prejudice

participants showing persistent stereotype-congruent responses. Presenting the finding as a general capability of controlled processes misaligns the evidence with the theoretical assertion, potentially misleading readers about the scope of the effect.

- Evidence:

- `results_and_discussion/exp_3` -- field `key_findings`: High-prejudice subjects did not differ in pejorative label generation and listed more negative thoughts, indicating no evidence of controlled inhibition for them.

- `general_discussion` -- field `results_stated_to_support_theory`: Study 3 is described as providing evidence that controlled processes can inhibit automatic processing, without qualifying that this effect was observed only for low-prejudice participants.

3. Internal Consistency

- **Experiment 2's introduction predicts both increased hostility ratings and a broader global negative evaluation of ambiguous behavior, but the results only report effects on hostility-related scales and provide no evidence regarding the predicted global negative evaluation.**

- Severity: **moderate**

- Justification: The paper's theoretical model expects automatic activation to influence both hostility-related and broader evaluative judgments. By failing to present data addressing the global evaluation prediction, the experiment leaves a gap between its stated hypothesis and the empirical evidence reported, reducing the coherence of the argument across sections.

- Evidence:

- `exp_introduction/exp_2` -- field `specific_predictions`: Prediction of a global negative evaluation beyond hostility is stated in the introduction.

- `results_and_discussion/exp_2` -- field `key_findings`: Findings report a significant Priming × Scale interaction for hostility-related scales only; no results are presented for a global negative evaluation.

4. Alternative Accounts

- **The tripartite model of attitudes, which treats stereotypes as the cognitive component of prejudice, is presented as an alternative theory but is never evaluated or discussed in the results or discussion sections.**

- Severity: **moderate**

- Justification: By introducing the tripartite model without later empirical comparison, the paper leaves a plausible competing explanation unexamined. This omission limits the ability to assess whether the dissociation of automatic and controlled processes uniquely accounts for the data, or whether a more parsimonious cognitive component view could also explain the findings.
- Evidence:
- `introduction` -- field `alternative_theories`: The introduction lists the "tripartite model of attitudes" as an alternative theory, contrasting it with the focal dissociation model.
- `general_discussion` -- field `alternative_accounts_addressed`: Only the "inevitability of prejudice" account is named and addressed; the tripartite model is absent.

5. Statistical Completeness

- **Experiment 2 does not report any main effects or interactions involving the between-subjects factor "task condition" (judgment vs. recognition vs. guess) despite it being a core predictor in the design.**
- Severity: **moderate**
- Justification: The experimental design includes a three-level between-subjects factor, which requires reporting its main effect and any interactions with other predictors (e.g., priming, prejudice). The absence of such reports indicates incomplete statistical reporting.

Evidence:

- `methods/exp_2` -- field `design_statement`: 3 (task condition: judgment vs recognition vs guess) between-subjects design
- `results_and_discussion/exp_2` -- field `key_findings`: Only descriptive statistics for the guess condition and ANOVAs that omit "task condition" are reported; no main effect or interaction involving task condition is presented.

Experiment 2 reports a significant three-way interaction (Prejudice Level × Priming × Replication) but provides no simple-effects or follow-up analyses to clarify the nature of this interaction.

- Severity: **moderate**
- Justification: When a higher-order interaction is statistically significant, standard practice is to conduct simple-effects or planned comparisons to interpret the interaction. The lack of such follow-up constitutes an omission in the statistical completeness of the report.
- Evidence:

- `results_and_discussion/exp_2` -- field `key_findings`: Finding: "Prejudice Level × Priming × Replication interaction, $F(1,70) = 4.69$, $p < .03$" reported without any subsequent simple-effects tests or decomposition.

6. Methodological Concerns

- **No power analysis or sample size justification is reported for any of the three experiments**
- Severity: **moderate**
- Justification: Without a priori power calculations, it is unclear whether the per-condition sample sizes (e.g., 21 high-prejudice vs 19 low-prejudice in exp 1) provide adequate statistical power to detect the hypothesized effects. This limits confidence in null findings and may inflate Type II error rates.

Evidence:

- `methods/exp_1` -- field `power_analysis`: power_analysis is null
- `methods/exp_2` -- field `power_analysis`: power_analysis is null
- `methods/exp_3` -- field `power_analysis`: power_analysis is null

Counterbalancing procedures are not described for any experiment that uses multiple stimulus conditions

- Severity: **moderate**
- Justification: Both exp 2 (varying priming proportion and stimulus replication) and exp 3 (thought-listing tasks) involve multiple conditions that could be order- or position-confounded. Absence of a reported counterbalancing scheme raises concerns about systematic bias.

Evidence:

- `methods/exp_1` -- field `counterbalancing`: counterbalancing: not reported
- `methods/exp_2` -- field `counterbalancing`: counterbalancing: not reported
- `methods/exp_3` -- field `counterbalancing`: counterbalancing: not reported

Preregistration status is not reported for any of the experiments

- Severity: **minor**
- Justification: Transparency about preregistration would help assess analytic flexibility. Its omission does not invalidate the studies but limits reproducibility checks.

Evidence:

- `methods/exp_1` -- field `preregistration_status`: preregistration_status: not reported
- `methods/exp_2` -- field `preregistration_status`: preregistration_status: not reported
- `methods/exp_3` -- field `preregistration_status`: preregistration_status: not reported

Exclusion criteria and participant recruitment details are insufficiently reported, especially in Experiment 1

- Severity: **moderate**
- Justification: Clear exclusion criteria are essential for reproducibility and for understanding sample composition. The lack of reported recruitment source and systematic exclusion rules in exp 1 hampers assessment of potential selection bias.

Evidence:

- `methods/exp_1` -- field `exclusions`: exclusions: [] (none reported)
- `methods/exp_1` -- field `recruitment_source`: recruitment_source: null
- `methods/exp_2` -- field `exclusions`: list of specific exclusions (e.g., dyslexia, failure to follow instructions) but no systematic criteria

Reliability estimates for coded subjective measures (e.g., thought-listing coding categories) are not provided

- Severity: **minor**
- Justification: Subjective coding without reported reliability (e.g., Cohen's κ) raises questions about measurement consistency, though it does not fundamentally undermine the reported effects.
- Evidence:
- `methods/exp_3` -- field `outcome_measures`: thought-listing coded into categories, but no inter-rater reliability reported

7. Statistical Robustness

- **Effect sizes and confidence intervals are not reported for any of the key ANOVA or t-test results, limiting assessment of the practical significance of findings.**
- Severity: **moderate**
- Justification: Without effect sizes or confidence intervals, readers cannot gauge the magnitude or reliability of the reported effects. This omission hampers the interpretation of

whether statistically significant findings are substantively meaningful, especially given the modest sample sizes.

Evidence:

- `results_and_discussion/exp_2` -- field `key_findings`: All reported statistics (e.g., $F(1,70) = 5.04$, $p < .03$; $t(64) = .68$, $p > .10$) lack accompanying effect size metrics (e.g., η^2, d) or confidence intervals.
- `results_and_discussion/exp_3` -- field `key_findings`: ANOVA and t-test results (e.g., $F(1,65) = 28.82$, $p < .0001$) are presented without effect sizes or CIs.

Multiple pairwise and higher-order ANOVA interactions are tested without any correction for familywise error rate, increasing the risk of Type I errors.

- Severity: **moderate**
- Justification: Testing many effects inflates the probability of false positives. The absence of any stated correction undermines confidence that the reported significant interactions reflect true effects rather than chance findings.

Evidence:

- `results_and_discussion/exp_2` -- field `key_findings`: Numerous interactions are examined (e.g., Prejudice × Priming × Replication, Prejudice × Scale Relatedness × Replication) with p-values reported as significant ($p < .03$, $p < .04$) but no mention of Bonferroni, Holm, or other correction procedures.
- `results_and_discussion/exp_3` -- field `key_findings`: Four separate interaction tests are reported (e.g., Prejudice × Valence, Prejudice × Type, three-way interaction) with $p < .0001$ or $p < .03$, yet no adjustment for multiple comparisons is described.

A marginal p-value (p > .10) is described as a non-significant result, but elsewhere p-values just above .05 (e.g., p = .06) are not explicitly qualified, leaving ambiguity about the treatment of marginal effects.

- Severity: **minor**
- Justification: Clear criteria for interpreting marginal significance are essential for transparent reporting. The inconsistent handling of p-values near conventional thresholds may lead readers to over- or under-estimate the evidence for certain effects.
- Evidence:
- `results_and_discussion/exp_3` -- field `key_findings`: The statement "$t(64) = .68$, $p > .10$" is presented without comment, while other results such as "$F(1,70) = 5.04$, $p < .03$" are highlighted as significant; the manuscript does not clarify how borderline p-values (e.g., .05

$< p < .10$) would be interpreted.